ORBIT BEHAVIOR

View online: https://www.construct.net/en/make-games/manuals/construct-3/behavior-reference/orbit

The **Orbit behavior** moves an object in a circle or ellipse around a point. The object's initial position is used as the point to orbit around.

Scripting

When using JavaScript or TypeScript coding, the features of this behavior can be accessed via the IOrbitBehaviorInstance script interface.

Orbit properties

Speed

The speed to orbit at, in degrees per second. Positive is clockwise and negative is anticlockwise.

Acceleration

The rate of change to the orbit speed, in degrees per second per second. Positive will accelerate in a clockwise direction and negative will accelerate in an anticlockwise direction.

Primary radius

The distance of the orbit from its center point, in pixels. For a circular orbit, ensure the primary and secondary radii are the same. For elliptical orbits, the primary radius is the one in the direction of the offset angle.

Secondary radius

The perpendicular distance of the orbit from its center point, in pixels. For a circular orbit, ensure the primary and secondary radii are the same. For elliptical orbits, the secondary radius is the one perpendicular to the offset angle.

Offset angle

For elliptical orbits, the rotation of the ellipse in degrees. For circular orbits, this does not affect the orbit path (since rotating a circle has no effect), but it changes the initial angle the orbit starts from.

Match rotation

If enabled, sets the object's angle to match the direction of travel in the orbit. If disabled the behavior only changes the object's position without affecting the angle.

Enabled

Whether the behavior is initially enabled or disabled. If disabled, it can be enabled at runtime using the *Set enabled* action.

Preview Paid plans only

Enable to run a preview of the behavior directly in the Layout View.

Orbit conditions

Is enabled

Test if the behavior is currently enabled.

Orbit actions

Pin

Unpin

Set another object as the location to orbit around, following the object if it moves. The *Unpin* action will stop following the object.

Set acceleration

Set enabled

Set match rotation

Set offset angle

Set radius

Set speed

Set the corresponding behavior properties. See *Orbit properties* above.

Set rotation

Set the current orbit position by its angle from the center point in degrees.

Set target

Set the center point of the orbit in layout co-ordinates.

Reset total rotation

Sets the counters for the TotalRotation and AbsoluteTotalRotation to 0

Orbit expressions

Acceleration

OffsetAngle

PrimaryRadius

SecondaryRadius

Speed

Return the corresponding behavior properties. See *Orbit properties* above.

DistanceToTarget

Return the distance from the object to the center point of the orbit, in pixels.

Rotation

Return the current position of the orbit as its rotation relative to the center point in degrees.

TargetX

TargetY

Return the current center point of the orbit in layout co-ordinates.

TotalRotation

Return the total rotation of the instance in degrees. This value does not wrap at 360 degrees. If the instance is rotating counter-clockwise then the value will decrease over time. This counter can be cleared using the *Reset total rotation* action.

TotalAbsoluteRotation

Return the total rotation of the instance in degrees, ignoring rotation direction. This expression is very similar to the *TotalRotation* expression, but rotation deltas are converted to absolute values. This means the counter will always increase even if the instance is rotating counter-clockwise. This counter can be cleared using the *Reset total rotation* action.